

	1	2	3	4	5	6
A	Power File: power.kicad_sch	Mechanical File: mechanical.kicad_sch				
B	Expansion bus connector File: expansion_bus.kicad_sch	Data bus buffers File: data_buffers.kicad_sch	Address bus buffers File: address_buffers.kicad_sch			
	Promulator File: promulator.kicad_sch	FPGA host interface File: fpga_host.kicad_sch	FPGA regen memory interface File: fpga_regen.kicad_sch			
C	Regen bus mux 1 File: regen_bus_mux_1.kicad_sch	Regen bus mux 2 File: regen_bus_mux_2.kicad_sch	Regen bus mux 3 File: regen_bus_mux_3.kicad_sch	Regen memory banks File: regen_mem.kicad_sch	Font RAM File: font_ram_2.kicad_sch	
D	Expansion ROM File: exp_rom.kicad_sch	RAMDAC File: ramdac.kicad_sch	I2C controller File: i2c.kicad_sch	Keyboard controller File: keyboard.kicad_sch		
	1	2	3	4	5	6

FREE FOR NON-COMMERCIAL USE

(C) TOM STOREY

Sheet: /

File: Video_Card.kicad_sch

Title: VGA VIDEO, AND PS/2 KEYBOARD AND MOUSE CARD

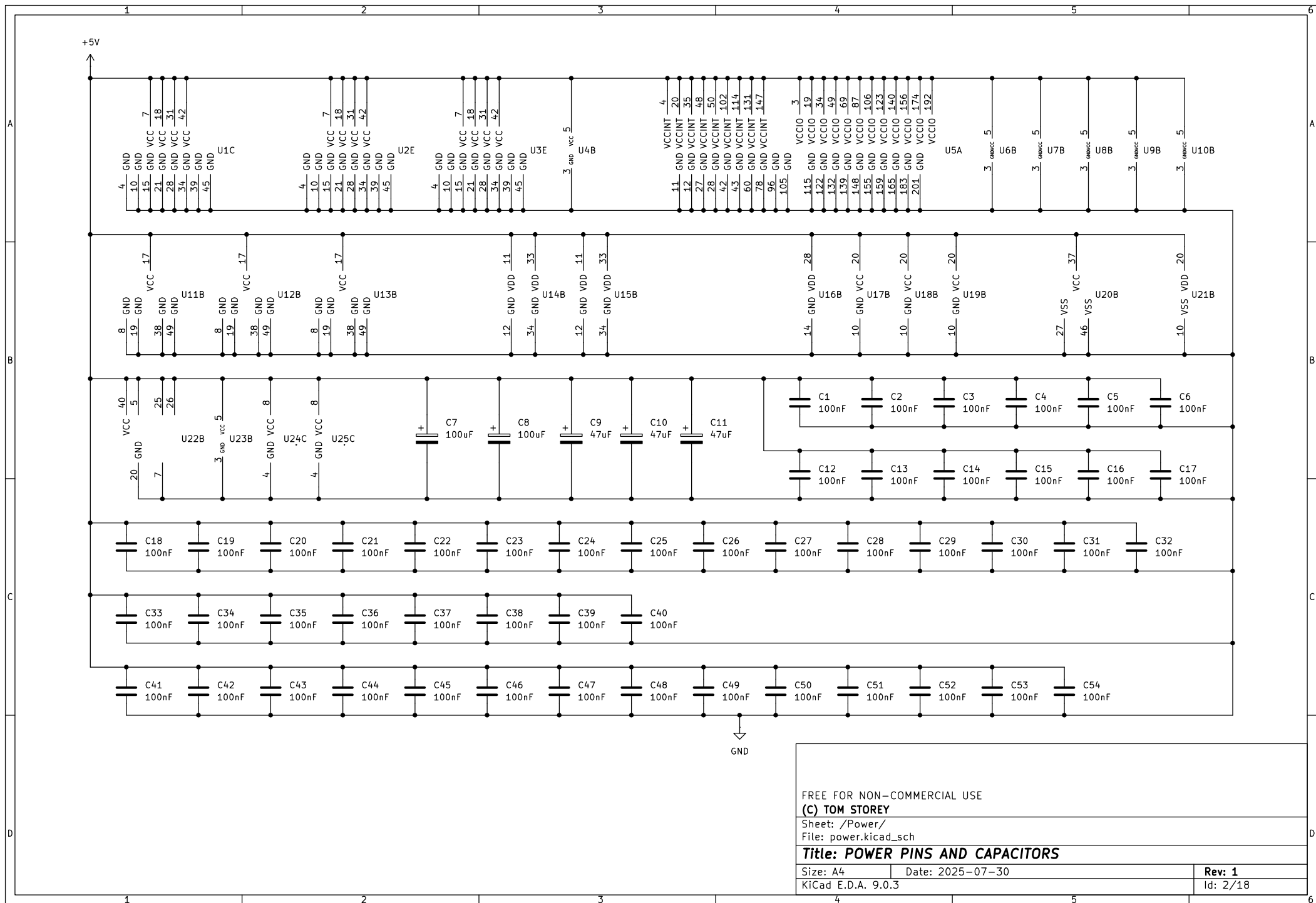
Size: A4

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(C) TOM STOREY

Sheet: /Power/

File: power.kicad_sch

Title: POWER PINS AND CAPACITORS

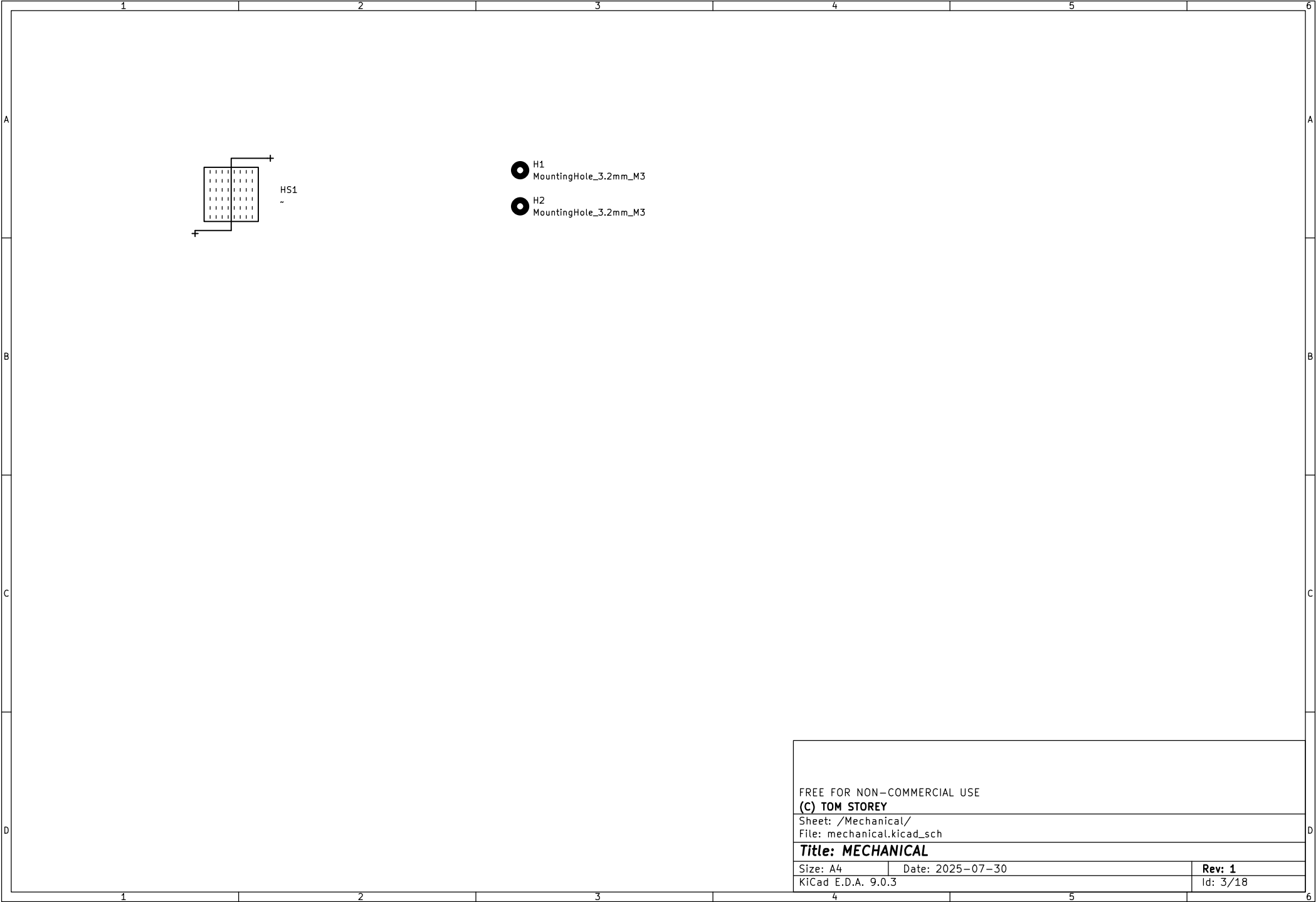
Size: A4

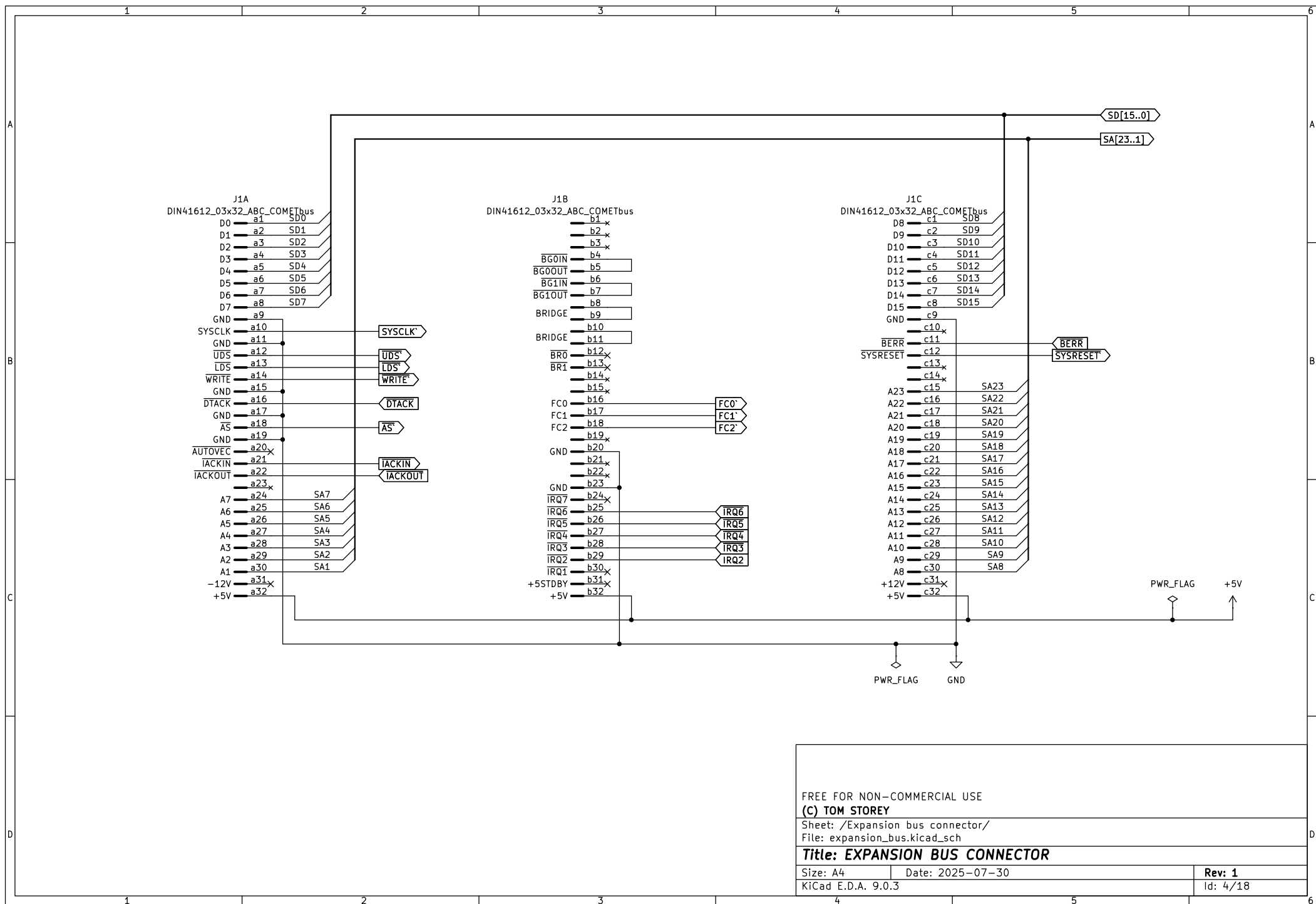
Date: 2025-07-30

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(C) TOM STOREY

Sheet: /Expansion bus connector/

File: expansion_bus.kicad_sch

Title: EXPANSION BUS CONNECTOR

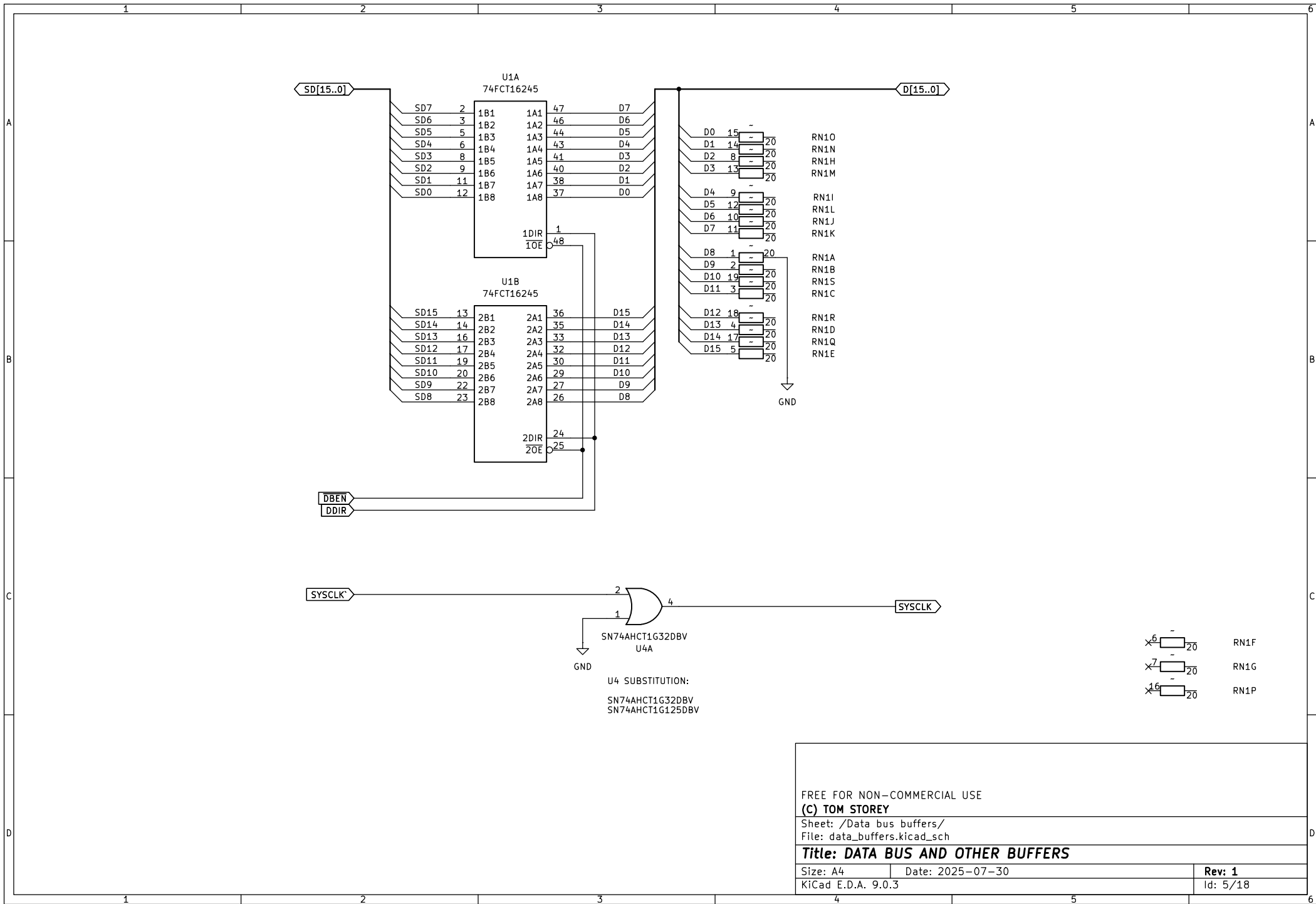
Size: A4

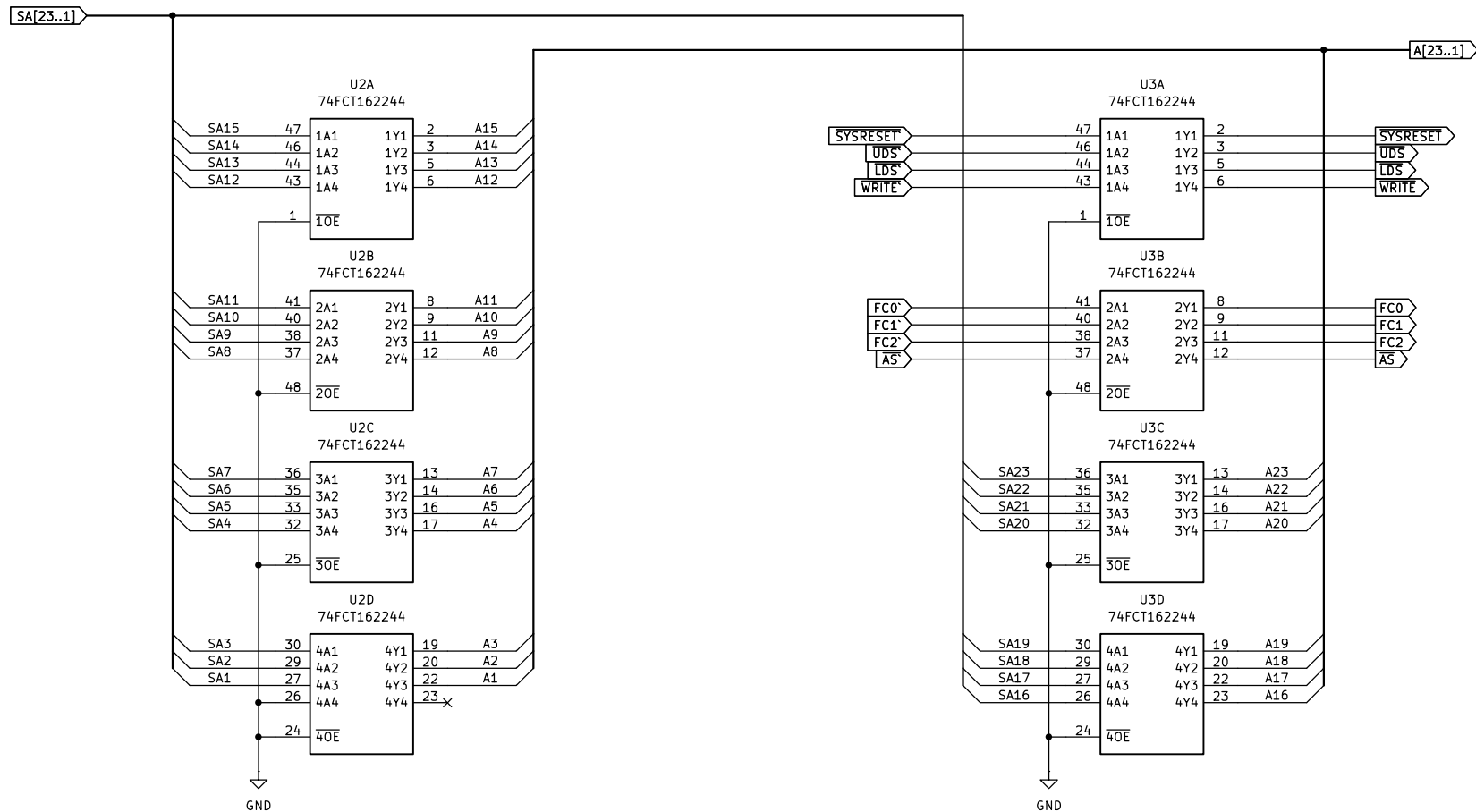
Date: 2025-07-30

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(C) TOM STOREY

Sheet: /Address bus buffers/

File: address_buffers.kicad_sch

Title: ADDRESS BUS AND CONTROL SIGNAL BUFFERS

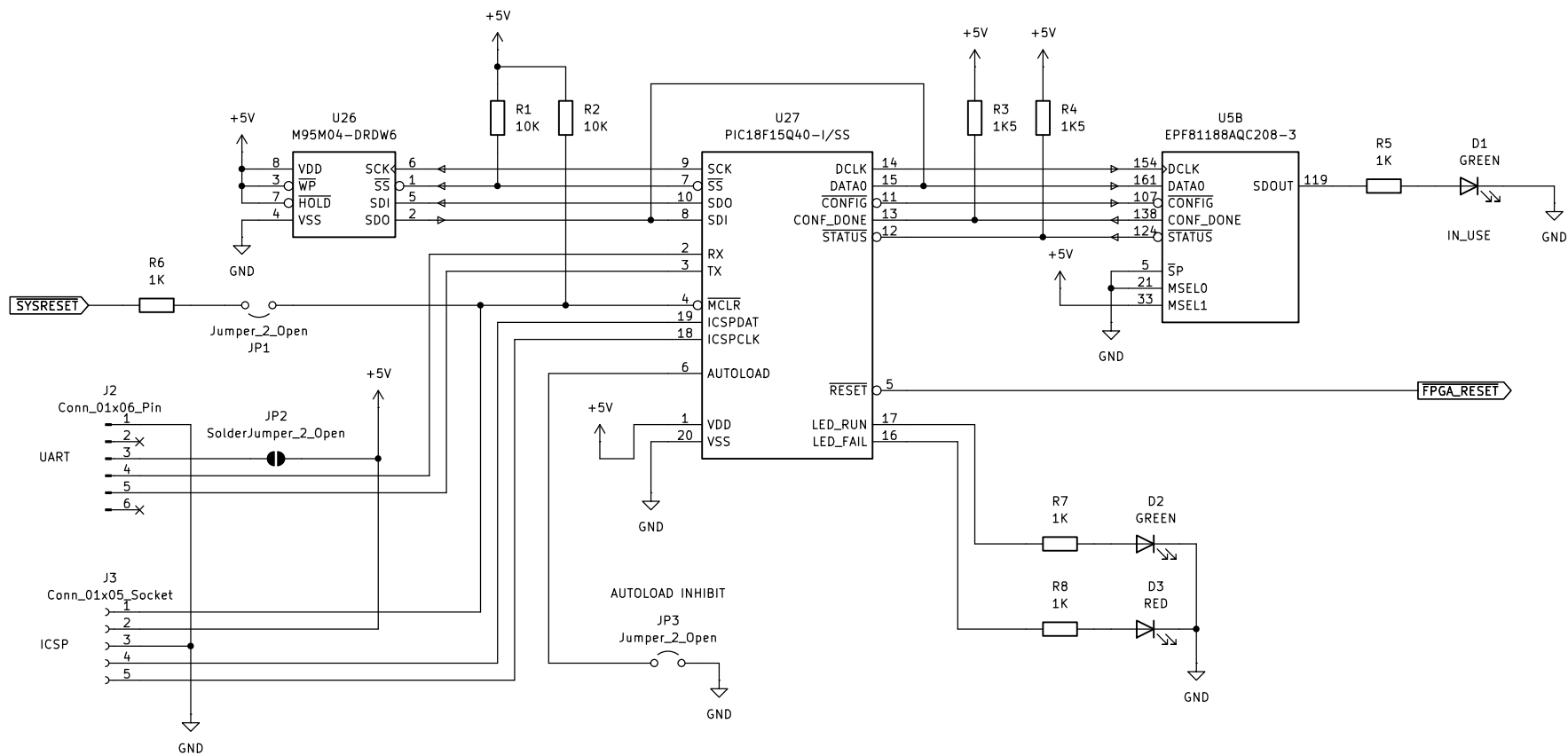
Size: A4

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$\overline{SP}$	MSEL1	MSEL0	MODE	
0	0	0	AS	ACTIVE SERIAL
0	0	1	RESERVED	RESERVED
0	1	0	PS	PASSIVE SERIAL **
0	1	1	RESERVED	RESERVED
1	0	0	APU	ACTIVE PARALLEL UP
1	0	1	PPS	PASSIVE PARALLEL SYNC
1	1	0	APD	ACTIVE PARALLEL DOWN
1	1	1	PPA	PASSIVE PARALLEL ASYNC

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(C) TOM STOREY

Sheet: /Promulator/

File: promulator.kicad_sch

Title: PROMULATOR - FPGA CONFIGURATION PROM EMULATOR

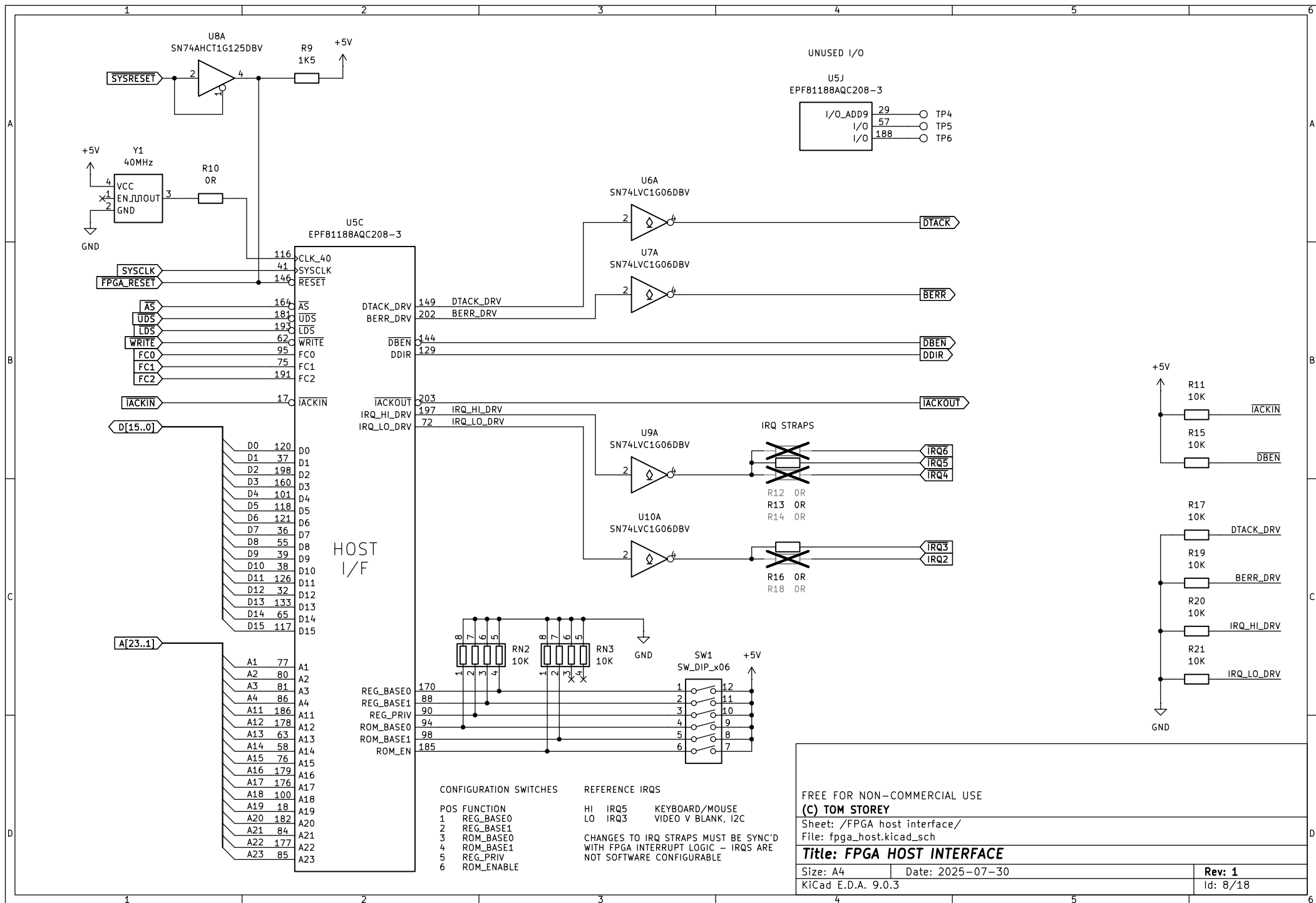
Size: A4

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(C) TOM STOREY

Sheet: /FPGA host interface/

File: fpga_host.kicad_sch

Title: FPGA HOST INTERFACE

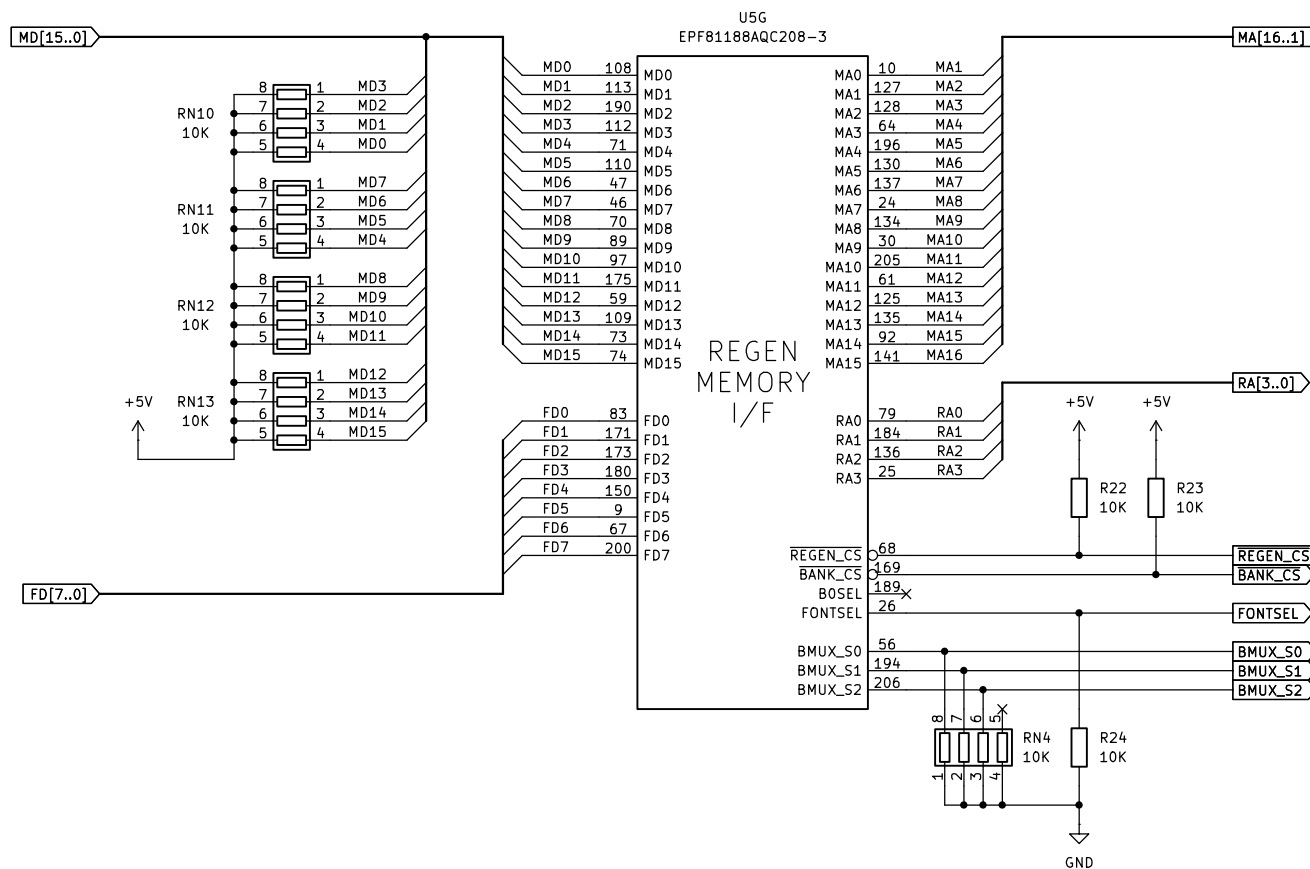
Size: A4

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MUX SELECT BITS

S	CONNECTION(S)
210	ALL DISCONNECTED
000	CPU ACCESSES BANK 0, CRTC DISCONNECTED
001	CRTC ACCESSES BANK 0, CPU DISCONNECTED
011	CPU ACCESSES BANK 0, CRTC ACCESSES BANK 1
110	CPU ACCESSES BANK 1, CRTC ACCESSES BANK 0
111	CPU ACCESSES BANK 1, CRTC ACCESSES BANK 0

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(C) TOM STOREY

Sheet: /FPGA regen memory interface/

File: fpga_regen.kicad_sch

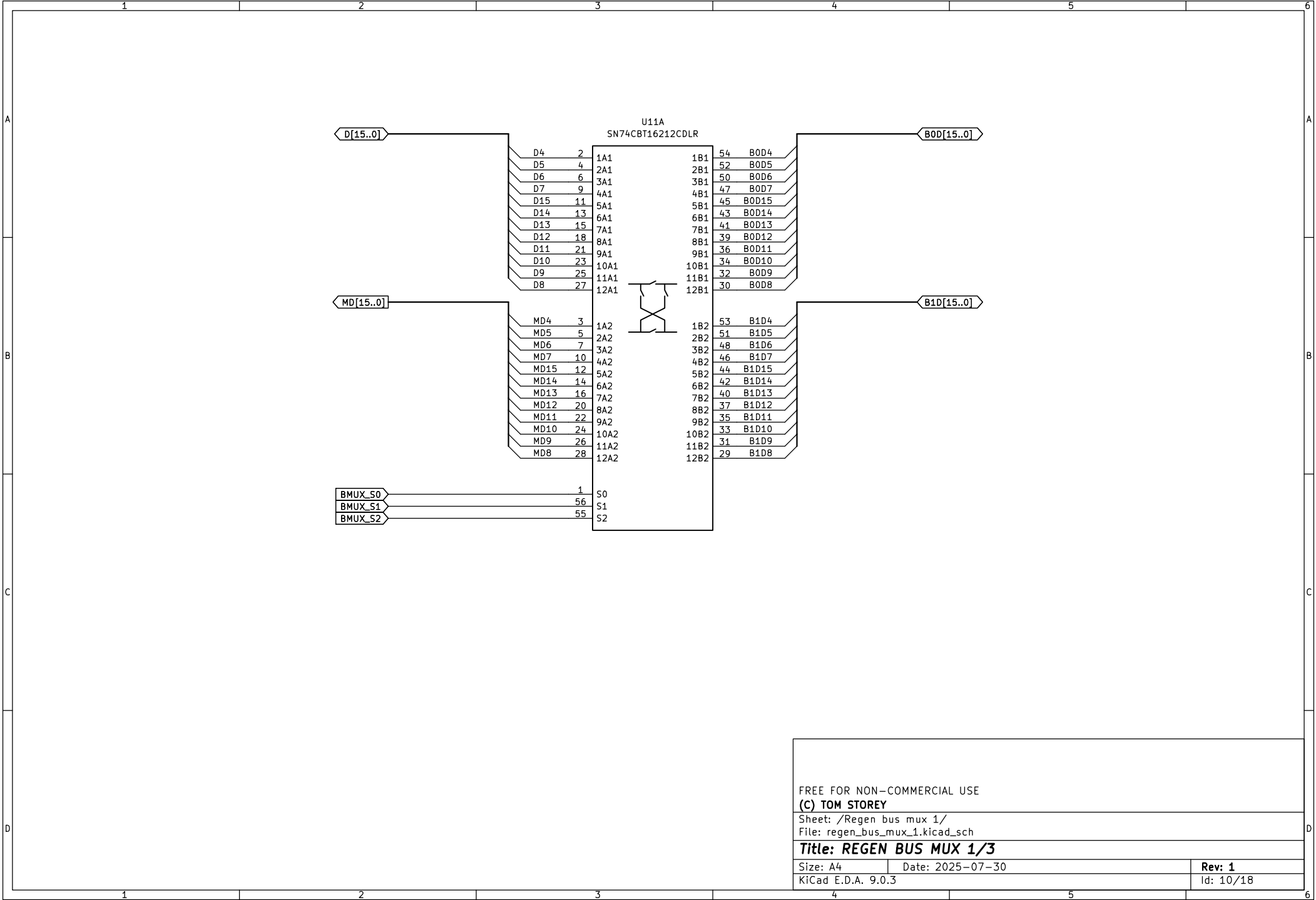
Title: **FPGA REGEN MEMORY INTERFACE**

Size: A4 Date: 2025-07-30

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(C) TOM STOREY

Sheet: /Regen bus mux 1/

File: regen_bus_mux_1.kicad_sch

Title: **REGEN BUS MUX 1/3**

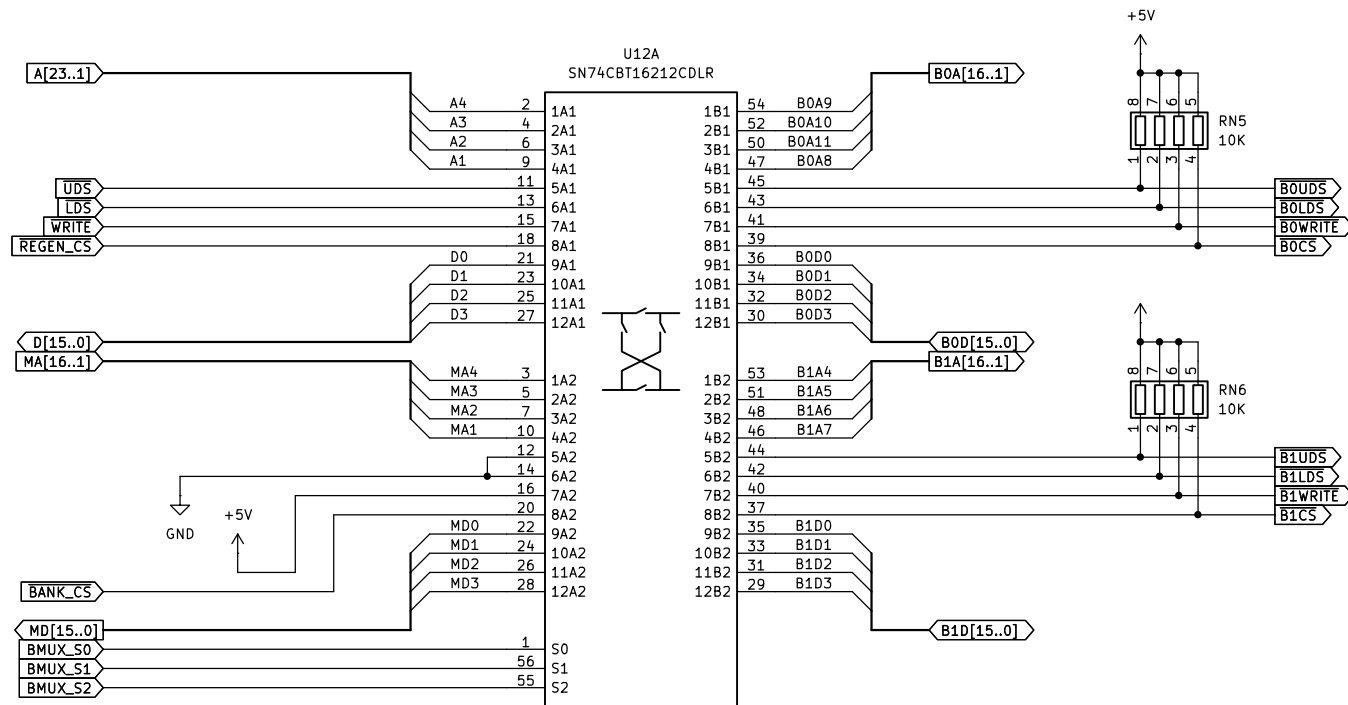
Size: A4

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REGEN MEMORY CONTROL SIGNALS

CRTC AND FORMATTER:

WRITE IS DRIVEN HIGH BECAUSE THE CRTC
ONLY EVER READS FROM REGEN MEMORY

$\overline{XDS}$ ARE DRIVEN LOW TO FORCE WORD READS
FROM REGEN MEMORY

CPU:

SIGNALS ARE DRIVEN ACCORDING TO THE
BUS CYCLE BEING EXECUTED

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(C) TOM STOREY

Sheet: /Regen bus mux 2/

File: regen_bus_mux_2.kicad_sch

Title: REGEN BUS MUX 2/3

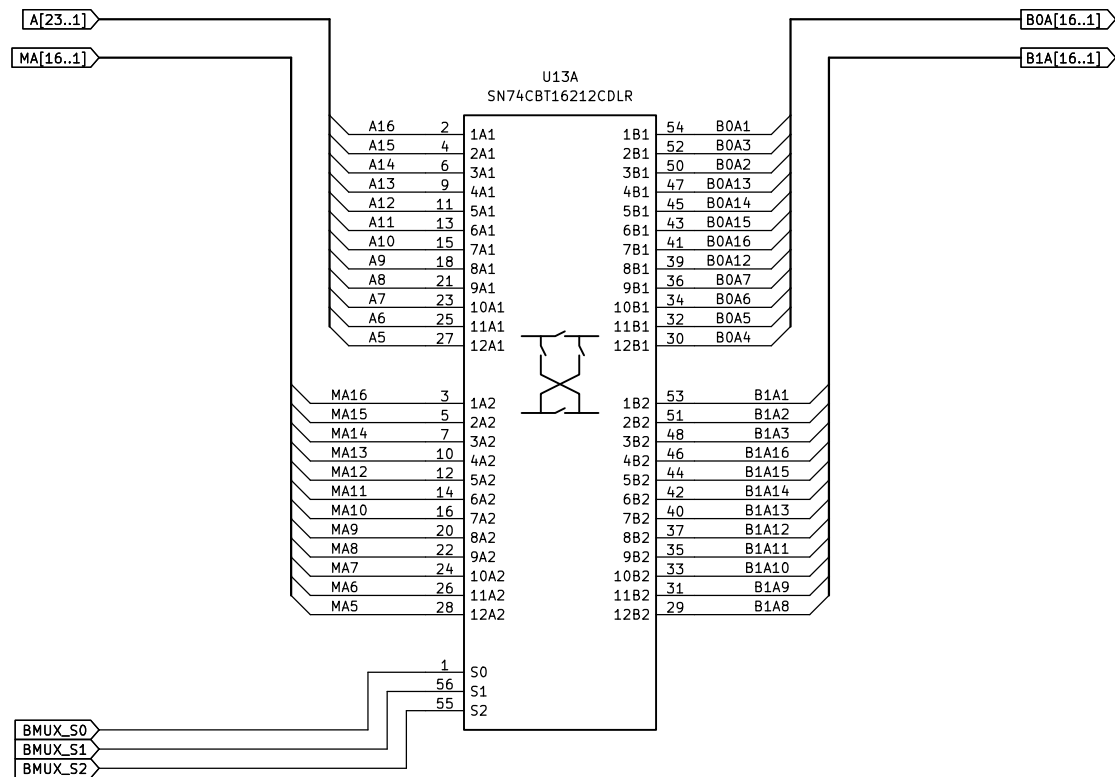
Size: A4

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(C) TOM STOREY

Sheet: /Regen bus mux 3/

File: regen_bus_mux_3.kicad_sch

Title: REGEN BUS MUX 3/3

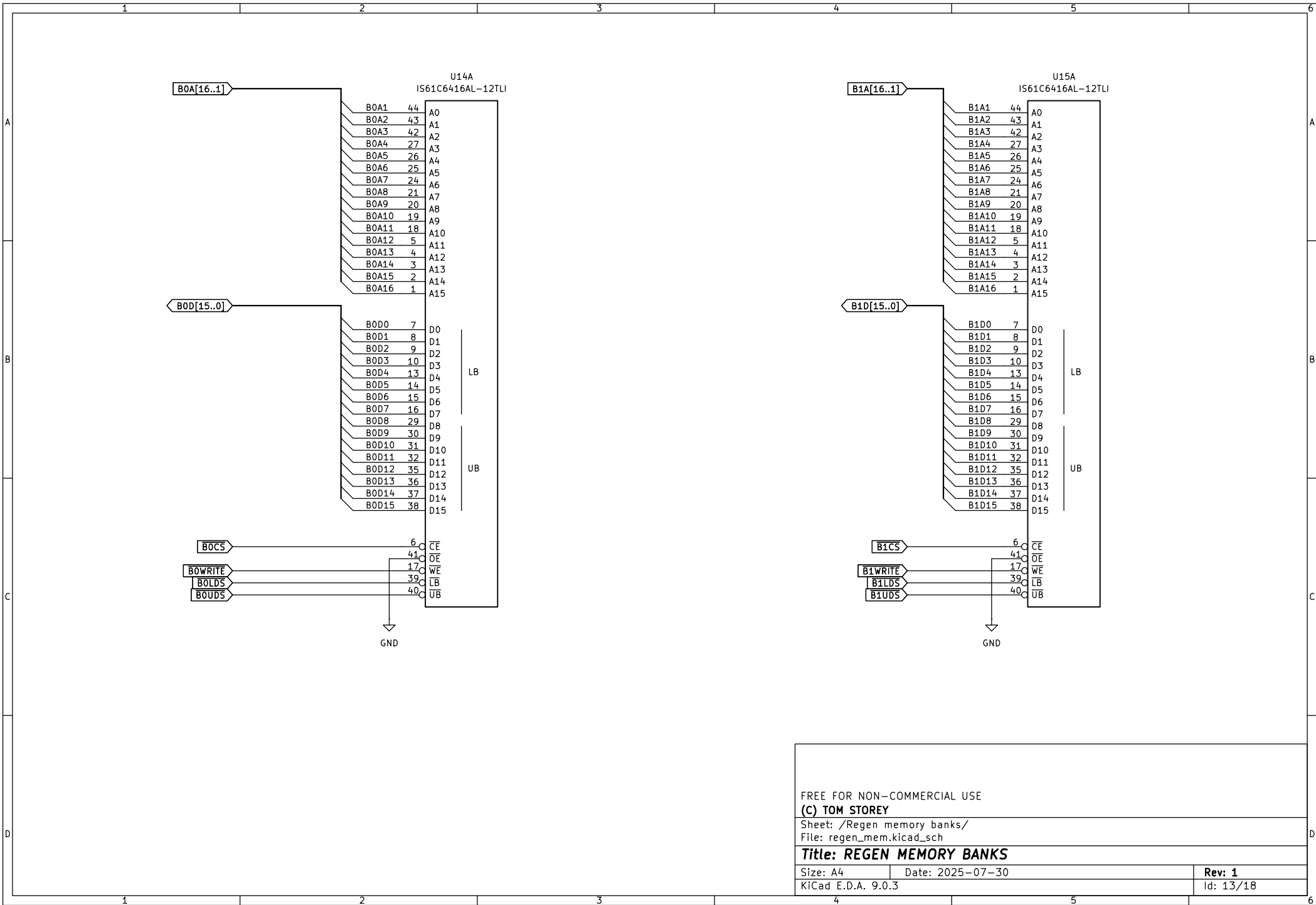
Size: A4

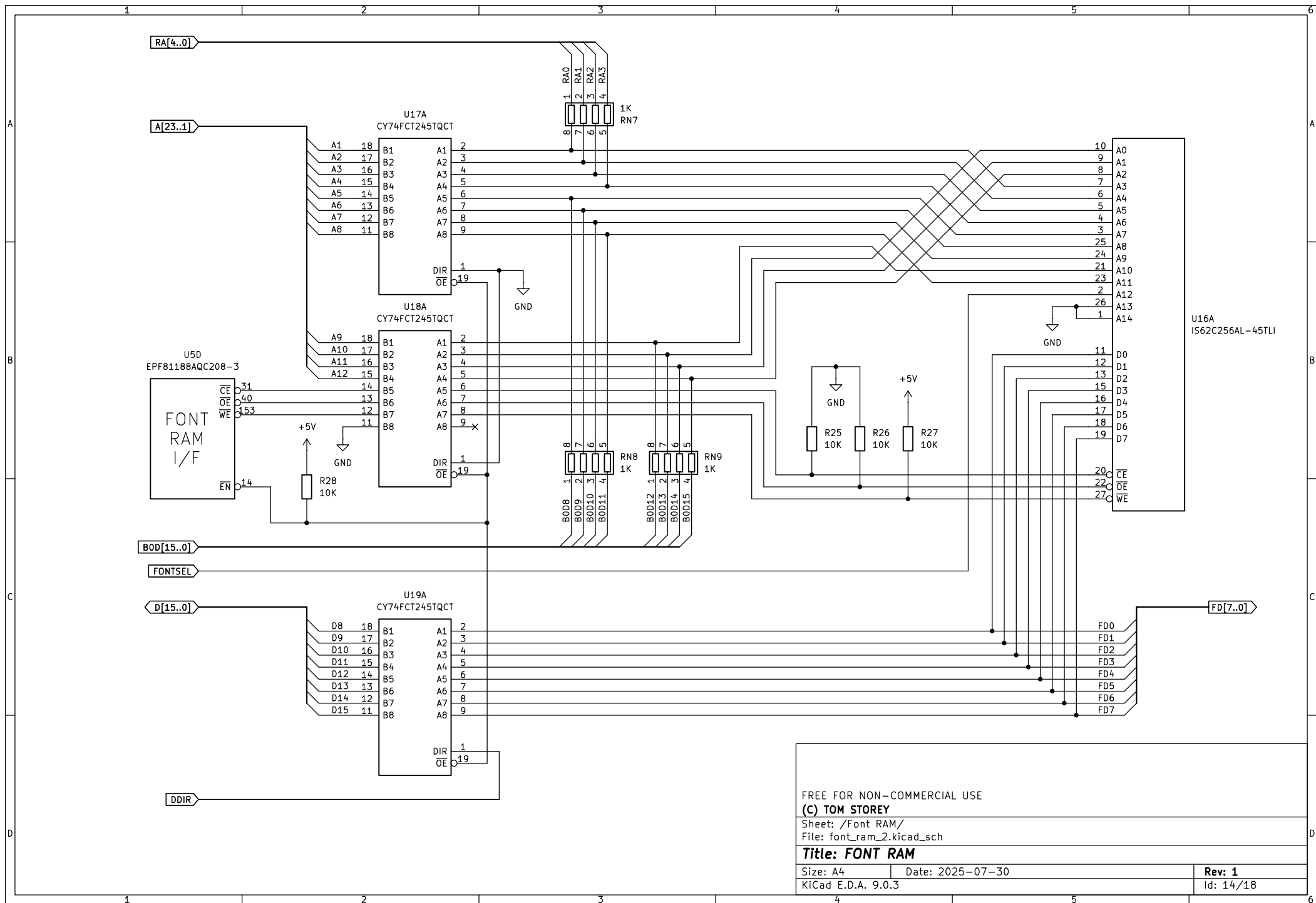
Date: 2025-07-30

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(C) TOM STOREY

Sheet: /Font RAM/

File: font_ram_2.kicad_sch

Title: FONT RAM

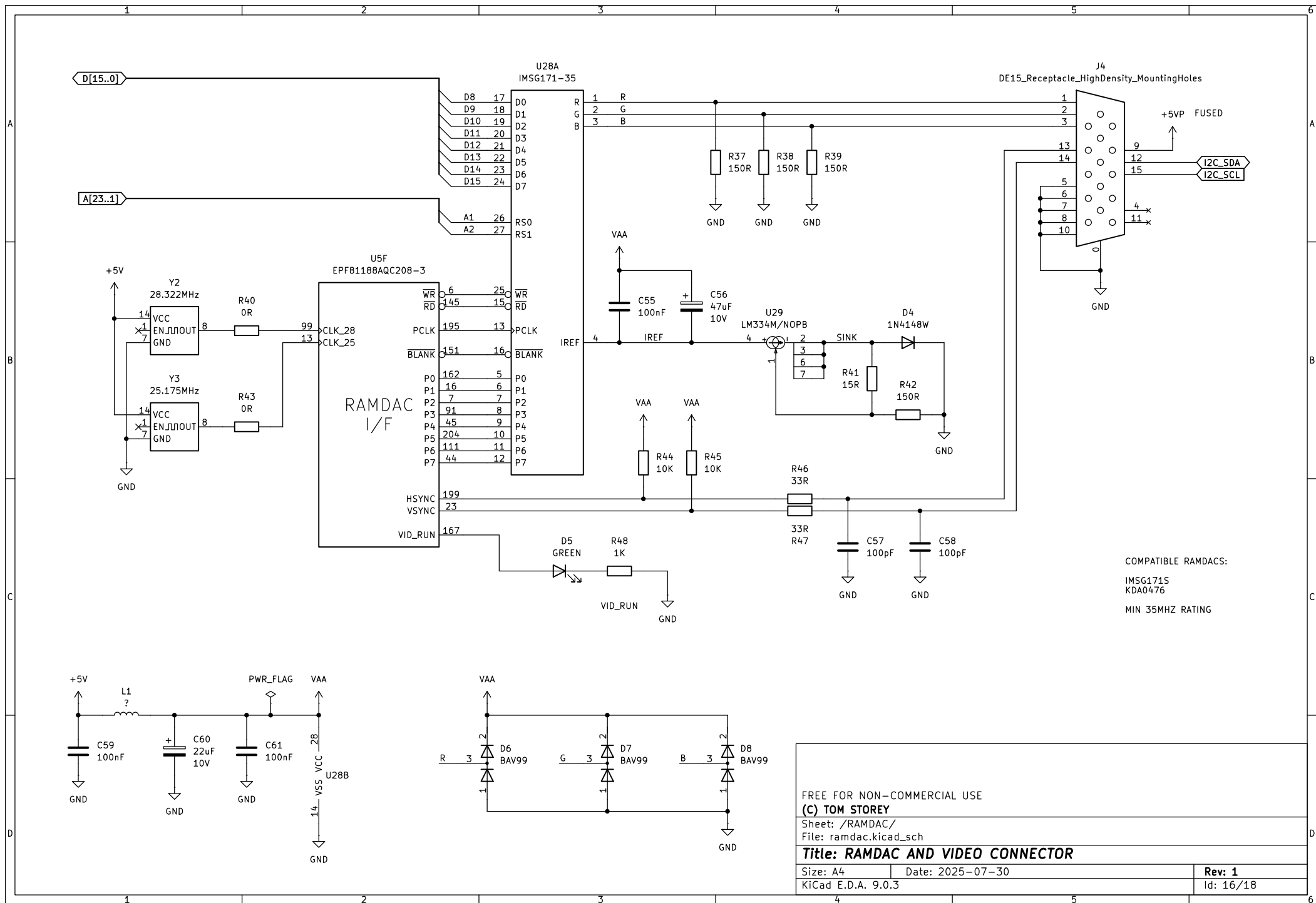
Size: A4

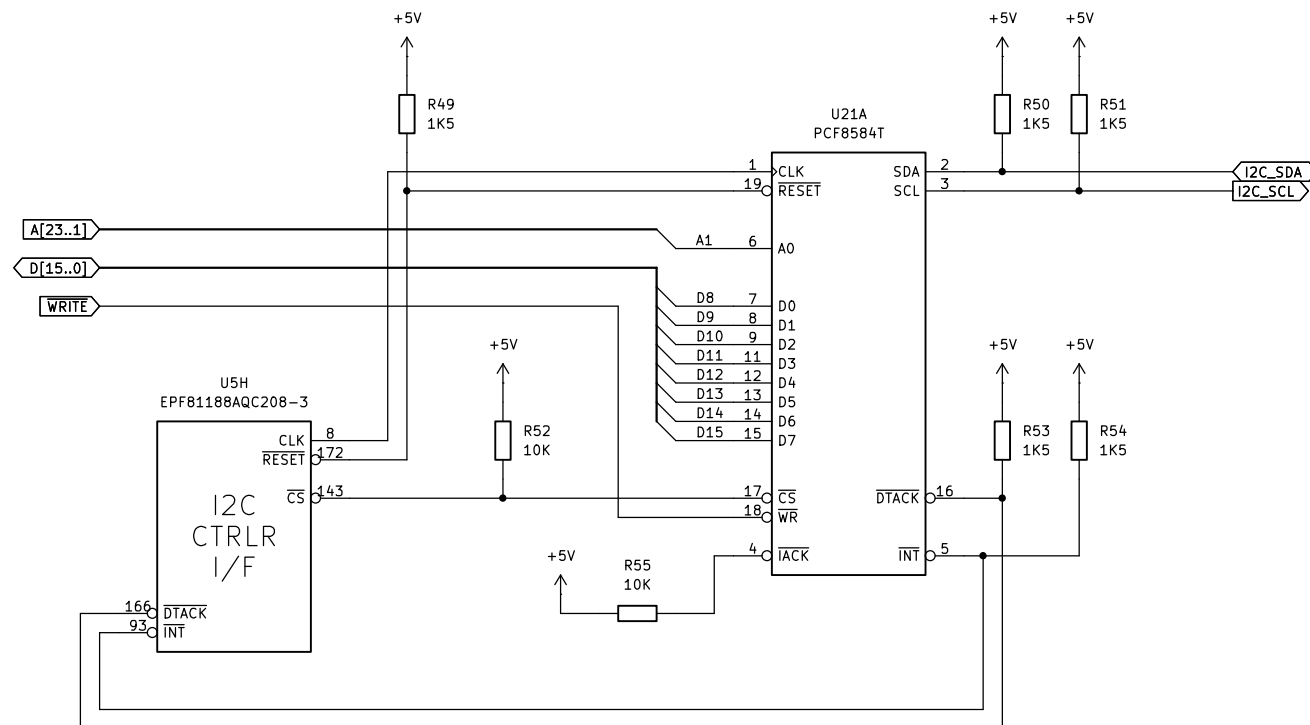
Date: 2025-07-30

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(C) TOM STOREY

Sheet: /I2C controller/

File: i2c.kicad_sch

Title: I2C CONTROLLER

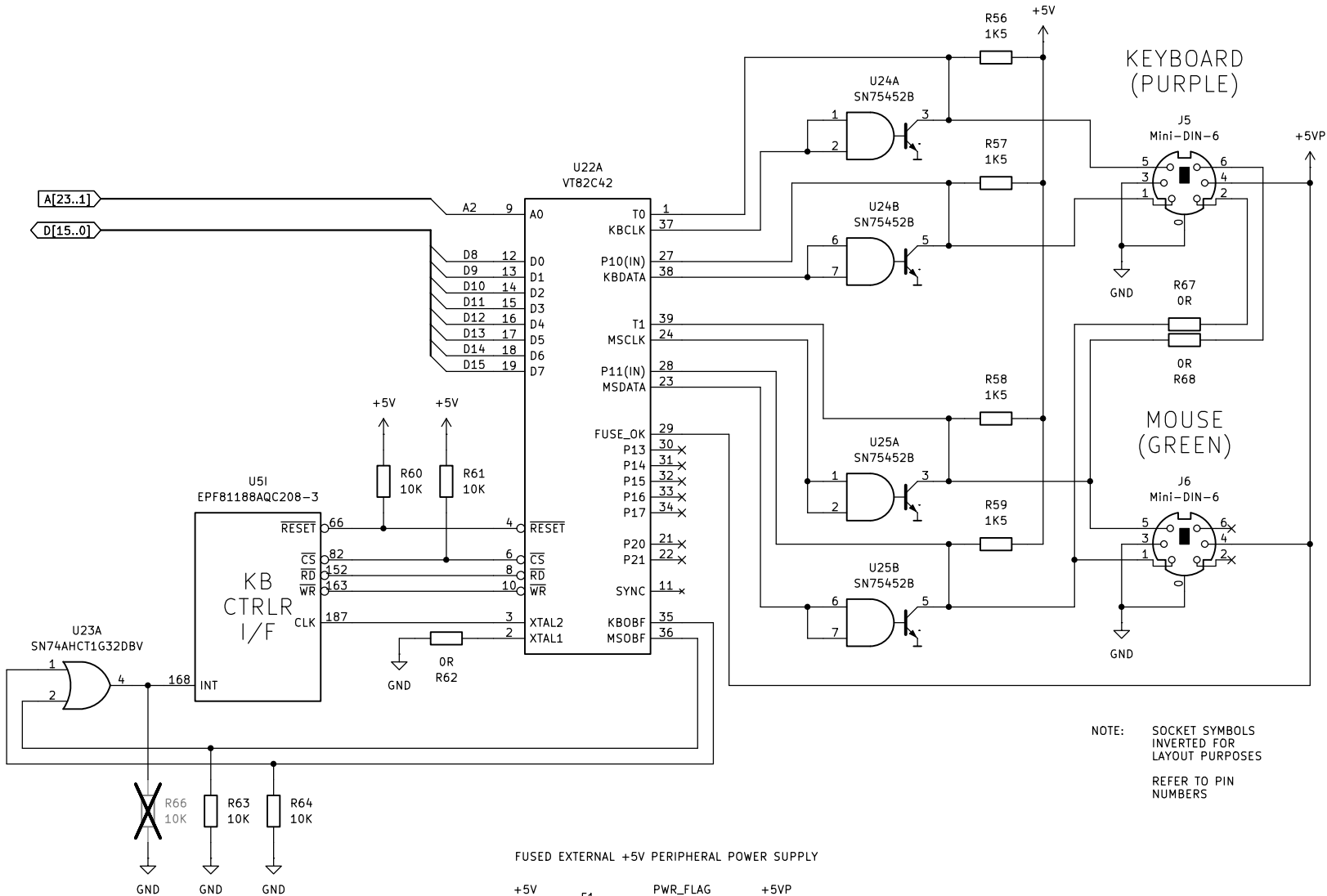
Size: A4

Date: 2025-07-30

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ALL COMPONENTS (EXCEPT +5VP POWER SUPPLY):
 NOPOP IF KEYBOARD/MOUSE NOT USED
 POPULATE R66 IF KEYBOARD/MOUSE NOT USED
 +5VP POWER SUPPLY COMPONENTS TO ALWAYS BE POPULATED

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(C) TOM STOREY

Sheet: /Keyboard controller/
 File: keyboard.kicad_sch

Title: PS/2 KEYBOARD AND MOUSE CONTROLLER

Size: A4 Date: 2025-07-30

KiCad E.D.A. 9.0.3

Rev: 1

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